PREREGISTRATION — EP-AGENT-001

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Cross-Framework ExecutionProof Portability and Integration Study

Field	Value
Study ID	EP-AGENT-001
Version	v0.3.5-DRAFT — FINAL PRE-FILING CANDIDATE LOCK (2026-10-16). Version history: see HISTORICAL / SUPERSEDED below and `CHANGELOG.md`
Date	2026-10-15
Owner / PI	Derek Hone, Remnant Fieldworks Inc.
System under integration	ExecutionProof Controlled Reference Verifier, tag `v0.1.2-reference`, commit `b03365d57880e896c35e72898f8335a1826afe74`, `POST /verify` (see `EXECUTIONPROOF_PIN.md`, final candidate pin; adoption via §18 amendment at filing)
Study type	Deterministic integration / portability study (RF internal)

```
DESIGN:                          COMPLETE
EXECUTIONPROOF PIN:              PASS
Q9:                              PASSED
Q8 TECHNICAL FREEZE READINESS:   PASS
Q8 FORMAL:                       OPEN — REQUIRES DEREK APPROVAL + PREREGISTRATION FILING
PREREGISTRATION:                 NOT FILED
EXPERIMENT:                      NOT EXECUTED
RESULTS:                         NONE
EXECUTION:                       NOT AUTHORIZED

DESIGN-SCOPE VERDICT:            GO WITH BOUNDARY
CURRENT EXECUTION ELIGIBILITY:   NOT YET ELIGIBLE (pending Q8 formal)
```

Lock definition (no other lock element exists). The preregistration lock consists of exactly: (1) the final SHA-256 hash of every manifest-tracked file, (2) `MANIFEST.sha256` and its own SHA-256, (3) the git commit hash containing them, (4) the UTC timestamp of that commit, and (5) the preregistration filing record. No digital signature, signing key, or cryptographic sign-off is part of the lock; none was preregistered.

The hashes currently in `MANIFEST.sha256` are a **FINAL PRE-FILING CANDIDATE LOCK** (not a final preregistration lock). They become the final lock only after Derek Hone approves the packet in writing and the preregistration is filed. Until then they do not constitute a preregistration lock.

HISTORICAL / SUPERSEDED status lines

Retained in chronological order (Rule 11). None of these is current status; the block above is operative.

Version	Superseded status line
v0.2	PREREGISTRATION DRAFT COMPLETE — HOLD FOR Q9 RECONCILIATION; NEW-RESEARCH GATE: PENDING Q8 + Q9
v0.3	Q8 TECHNICAL FREEZE READINESS: FAIL (checklist item 7 BLOCKED)
v0.3.1	Q9 PASSED; NEW-RESEARCH GATE: PENDING Q8 (item 7 still BLOCKED)

Version	Superseded status line
v0.3.2	Candidate pin `v0.1.1-reference` @ `279bcb2669e4602ae25e0d745d827a1c8514265e`; item 7 OPEN
v0.3.3	Final pin `v0.1.2-reference`; item 7 PASS; Q8 technical PASS; status HOLD FOR DEREK APPROVAL + FILING
v0.3.4	Status-line sync across packet documents (superseded values kept inline)

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1. Primary research question (frozen)

> **RQ:** Can selected AI-agent frameworks integrate with the same ExecutionProof Request Contract and verification boundary while preserving the same core ALLOW / HOLD / DENY semantics and ProofRecord linkage without modification of the ExecutionProof core decision engine?

2. Commercial question (separate — not a scientific endpoint)

> **CQ:** Can an implementation partner retain its preferred agent framework while ExecutionProof remains an independent pre-execution verification boundary?

The CQ is recorded for business context only. No endpoint, threshold, or classification in this preregistration answers the CQ. Scientific results must be reported before, and separately from, any commercial interpretation (see `CLAIM_BOUNDARY.md` §4).

3. Mandatory non-interference language

> "EP-AGENT-001 measures integration portability of the ExecutionProof decision path across selected agent frameworks under benign, policy-driven fixtures. It makes no claim about adversarial robustness, multi-agent authorization, or independent reproducibility, which are the subject of independent university projects."
>
> "University Non-Interference Boundary: This study is intentionally limited to framework portability, request normalization, verification-interface integration, decision propagation, ProofRecord linkage, and engineering integration characteristics. It does not test or resolve active University of Dayton or University of Washington adversarial, multi-agent authorization, or independent reproducibility research questions."

Source: Master University Non-Interference + Prior-Work Audit (2026-10-15), Section I — **DESIGN-SCOPE VERDICT: GO WITH BOUNDARY**. This is a design-scope verdict only. Current execution eligibility: **NOT YET ELIGIBLE** (new-research gate pending Q8 formal; Q9 PASSED; Q8 technical freeze readiness PASS; see `UNIVERSITY_NON_INTERFERENCE.md` §6).

No university role. No University of Dayton or University of Washington student, team, or faculty member has any role in EP-AGENT-001 — not consultant, co-investigator, participant, reviewer, or executor. No EP-AGENT-001 fixtures, expected outcomes, answer keys, or results are shared with any university team while its reserved project is active.

Interaction pattern (v1, fixed). One agent → one proposed consequential action → ExecutionProof `/verify` → simulated downstream boundary. Exactly one tool call per run; single-agent parallel tool calls are excluded.

4. System under integration (fixed)

```
Request Contract
→ Authority Engine
```

```
→ Evidence Engine
→ Constraint Engine
→ Control Engine
→ ProofRecord™
→ Execution Gateway
```

Core invariants assumed by the expected outcomes (they are the reference semantics, not hypotheses about robustness):

1. incomplete contract → HOLD
2. Authority FAIL → DENY
3. Authority PASS + Evidence FAIL → HOLD
4. Authority PASS + Constraint FAIL → HOLD
5. all required checks PASS → ALLOW
6. HOLD/DENY may not cross the controlled execution boundary
7. no cached permission assumption
8. current state evaluated at time of request
9. decision provenance preserved

Interfaces used: `/verify` (decision), `/proof/{id}` (ProofRecord retrieval). Policy endpoints (`/policies/authority`, `/policies/constraints`, `/policies/evidence-verifiers`, `/evidence-schemas`, `/policies/rails`) are used only to load the fixed fixture policy before execution.

5. Hypotheses

ID	Hypothesis	Failure condition (any one falsifies)
H1 — Integration	Each of the four frameworks can submit the frozen Request Contract to `/verify` through a thin adapter, for all five benign cases, without modification of ExecutionProof core code.	Any framework cannot complete integration for ≥1 case without (a) modifying ExecutionProof core code, or (b) altering the semantics of a frozen Request Contract field.
H2 — Decision fidelity	For every framework × case cell, the verdict observed by the adapter equals the frozen expected verdict in `EXPECTED_RESULTS.json`.	Any cell where observed verdict ≠ expected verdict (exact string match; no tolerance).
H3 — Core semantic invariance	For each benign case, the verdicts returned for the four frameworks are identical, and the four normalized contracts are semantically identical on all canonical fields (all fields except `request_id`, `framework_id`, `nonce`, `requested_at`).	Any case where ≥1 framework's verdict differs from the others, or where canonical-field values or `action_hash` differ between frameworks.
H4 — ProofRecord linkage & verdict return	For every cell, `/verify` returns a ProofRecord reference that resolves via `/proof/{id}`, and the adapter returns both the verdict and the reference, unchanged, into the framework-native tool result of the same single agent.	Any cell where the reference is absent, does not resolve, or is not present unchanged in the framework-native result; or where the verdict string is altered on return.

H4 is a benign verdict-return and linkage measurement within one agent's single tool call. "Verdict return" never means authority propagation between agents (UW P2-reserved). It is **not** a test of ProofRecord integrity under tamper/forgery (UD-reserved) and **not** a containment-resistance test.

6. Primary endpoints

Notation: F = {openai_agents, langgraph, crewai, maf} (|F| = 4); C = {B1, B2, B3, B4, B5} (|C| = 5); cell (f, c); I(·) = indicator (1 if true, else 0); v(f,c) = observed verdict; e(c) = expected verdict.

ID	Endpoint	Exact formula	PASS threshold
P1	Integration completion rate	P1 = Σ_f Σ_c I(integration_complete(f,c)) / 20	20/20 (= 1.0)
P2	Decision fidelity	P2 = Σ_f Σ_c I(v(f,c) = e(c)) / 20	1.0 (100%)
P3	Core semantic invariance (per case)	P3_c = |{f ∈ F : v(f,c) = v_mode(c)}| / 4, where v_mode(c) is the most frequent verdict for case c (ties → P3_c = 0.5 or lower as computed)	P3_c = 1.0 for every c ∈ C
P4	ProofRecord linkage	P4 = Σ_f Σ_c I(proof_ref(f,c) returned ∧ resolves via /proof/{id} ∧ present in framework-native result) / 20	20/20
P5	Core modification required	P5 = boolean: was any change to ExecutionProof core code, core configuration semantics, or decision logic required for any framework?	false

`integration_complete(f,c)` = the adapter (i) normalized the fixture into a contract that passes the frozen schema, (ii) submitted it to `/verify` from inside the framework's native single-agent tool-call path, (iii) received a well-formed decision response, and (iv) returned that decision to the framework runtime — all without manual intervention during the run.

7. Secondary endpoints (engineering metrics — descriptive only)

ID	Metric	Measurement method
S1	Adapter LOC	Non-blank, non-comment, non-docstring Python lines in `adapters/<f>/adapter.py` + `normalizer.py`, counted by a frozen counting script at freeze; framework-specific lines reported separately from shared logic
S2	Setup burden	Number of documented install/config steps; wall-clock time from clean venv to first successful `/verify` call (time-to-first-verify)
S3	Dependency burden	Count of packages added to the clean lockfile by installing the framework; installed size (MB) of the framework's site-packages delta
S4	Latency — normalization	Time inside `normalize_to_request_contract` (ns resolution, `time.perf_counter_ns`)
S5	Latency — adapter round trip	Time from adapter entry (tool invocation received) to verdict returned to framework, including `/verify` round trip; median and p95 per cell; labelled **RF internal measurement**
S6	Integration friction	Count and description of workarounds, framework-specific code paths, and deviations; qualitative log per framework
S7	Configuration surface	Number of framework-specific configuration keys/objects required by the adapter

Secondary endpoints never alter the PASS / PARTIAL / FAIL / INCONCLUSIVE classification.

8. Benign test matrix

All cases use the single action archetype `SUBMIT_EXTERNAL_ACTION` against a mock target. Each case is applied identically to all four frameworks (20 cells). Full detail in `TEST_MATRIX.md`.

Case	Name	Description	Fixture	Expected	Rationale (core invariant)	University-interference check
B1	ALLOW_COMPLETE	Complete contract; authorized actor; verified evidence present; amount 50 ≤ limit 100	`fixtures/benign/B1_allow_complete.json`	**ALLOW**	All required checks PASS → ALLOW	PASS — well-formed, benign, no attack vector

Case	Name	Description	Fixture	Expected	Rationale (core invariant)	University-interference check
B2	HOLD_INCOMPLETE_CONTRACT	Exactly one required benign field absent: `current_state_reference` is omitted, while the fixture policy (`contract_completeness_requirement`) requires it for `SUBMIT_EXTERNAL_ACTION`. Everything else identical to B1 (evidence verified, authorized actor, amount 50). After normalization the field is `null`; `null` under a policy that requires the field = incomplete → HOLD	`fixtures/benign/B2_hold_incomplete_contract.json`	**HOLD**	Incomplete contract → HOLD	PASS — schema-valid, benign omission; not malformed, not adversarial
B3	DENY_NO_AUTHORITY	`actor_id = test-actor-unauthorized` has no authority grant for `SUBMIT_EXTERNAL_ACTION` in the static fixture policy	`fixtures/benign/B3_deny_no_authority.json`	**DENY**	Authority FAIL → DENY	PASS — static "role not granted" policy condition; no revocation, no forgery, no stale authority
B4	HOLD_EVIDENCE_ABSENT	Required benign evidence simply absent: `evidence_references = []` and no evidence record, while the policy requires ≥1 verified reference. Everything else identical to B1 (contract complete incl. `current_state_reference`)	`fixtures/benign/B4_hold_evidence_absent.json`	**HOLD**	Authority PASS + Evidence FAIL → HOLD	PASS — evidence simply absent; no forged, stale, replayed, malformed, manipulated, or adversarial evidence
B5	HOLD_CONSTRAINT_EXCEEDED	`parameters.amount = 101`; fixture policy constraint `amount <= 100`	`fixtures/benign/B5_hold_constraint_exceeded.json`	**HOLD**	Authority PASS + Constraint FAIL → HOLD	PASS — ordinary policy limit; not a parameter mutation or post-verification drift

9. Classification rules (summary — full text in `PASS_FAIL_RULES.md`)

Class	Exact condition
PASS	P1 = 20/20 ∧ P2 = 1.0 ∧ P3_c = 1.0 ∀c ∧ P4 = 20/20 ∧ P5 = false
PARTIAL	Not PASS and not FAIL, and ≥1 framework achieves 5/5 complete, faithful, linked cells with no core modification, while ≥1 framework fails ≥1 cell or requires a material workaround (a framework-specific change outside the adapter, or a documented deviation affecting the contract path)
FAIL	P5 = true; or ≥2 frameworks cannot use the same frozen contract without changing core semantics; or no framework achieves 5/5 complete, faithful, linked cells
INCONCLUSIVE	Only the preregistered invalidation conditions: upstream framework defect, environment failure, dependency outage, corrupted artifacts, incomplete execution caused by non-study infrastructure. Never used to rescue a negative result.

Precedence: FAIL is evaluated first, then INCONCLUSIVE (only if an invalidation condition is documented and the negative outcome is attributable to it), then PASS, then PARTIAL.

10. Frameworks (exact versions; pinned at freeze)

Framework	Package	Version	License	Integration point (single-agent only)
OpenAI Agents SDK	`openai-agents`	0.24.0	MIT	Function tool wrapping the verify call; handoffs excluded
LangGraph	`langgraph`	1.2.15	MIT	Tool executed via `ToolNode`; single-agent graph only
CrewAI	`crewai`	1.15.28	MIT	`BaseTool` subclass on a single agent, single task; delegation disabled
Microsoft Agent Framework	`agent-framework`	1.22.0	MIT	`FunctionMiddleware` around function tools (`call_next` only on ALLOW)

Versions, release dates, licenses, and sources were verified against PyPI, GitHub releases, and official docs on 2026-10-15 (`FRAMEWORK_SELECTION.md` §2a). Framework-native multi-agent features (handoffs, crew delegation, multi-agent graphs, sub-agents, A2A) are **excluded** and must be disabled or unused. **Single-agent parallel tool calls are also excluded in v1**: each run makes exactly one tool call (e.g. `parallel_tool_calls=False` where the framework exposes it). Google ADK 2.12.0 is **not excluded**: it can be configured strictly single-agent, single-tool without A2A or transfer (`FRAMEWORK_SELECTION.md` §4). Decision recorded 2026-10-15 (Derek Hone): **KEEP v1 AT FOUR FRAMEWORKS — ADK is NOT added to v1.** ADK status: ELIGIBLE FUTURE EXTENSION / NOT INCLUDED IN v1 / NO CHANGE TO 20-CELL MATRIX. v1 = `openai-agents==0.24.0`, `langgraph==1.2.15`, `crewai==1.15.28`, `agent-framework==1.22.0`.

11. Environment (summary — full text in `ENVIRONMENT.md`)

Linux (Ubuntu 22.04 LTS+) or macOS 14+; x86_64 or arm64; Python 3.11.x; pip with `requirements.lock`; ExecutionProof local/sandbox instance pinned by version/commit at freeze; no production credentials, no live customer systems, mock downstream targets only; no secrets in environment variables.

12. Analysis plan (summary — full text in `ANALYSIS_PLAN.md`)

Deterministic, descriptive reporting. Binary endpoints as exact counts and percentages. Decision fidelity is exact agreement with no tolerance. Latency: 5 warmup calls discarded, then 30 measured calls per case per framework; report median and p95; optional bootstrap 95% CI for the median (1,000 resamples, seed frozen at freeze). Raw values are reported alongside every summary. No inferential statistics are manufactured for trivial technical variability.

13. Claim ceiling

> **Maximum allowed successful claim:** "In an RF-run internal study, the ExecutionProof `/verify` decision path was integrated with four selected agent frameworks (OpenAI Agents SDK 0.24.0, LangGraph 1.2.15, CrewAI 1.15.28, Microsoft Agent Framework 1.22.0) through thin single-agent adapters using one frozen Request Contract. Across five benign, policy-driven fixtures, all 20 framework × case cells returned the preregistered ALLOW / HOLD / DENY verdict with ProofRecord linkage, without modification of the ExecutionProof core decision engine."

Required qualifiers and prohibited claims: `CLAIM_BOUNDARY.md`. Claims remain narrower than evidence.

14. Exclusions (UNIVERSITY-RESERVED — not implemented, not executed)

Excluded item	Reserved for
Agent-to-agent delegation / handoff	UW P2
Multi-agent authority / multi-hop delegation	UW P2
Collusion / confused deputy	UW P2
Shared authority, inherited permissions, authority propagation between agents	UW P2
Parallel agents / multi-agent chains / orchestration	UW P2
Single-agent parallel tool calls (excluded from v1 to keep one agent → one action)	v1 scope rule
Bypassing the adapter / direct unauthorized tool invocation	UD / UW P1
Replay through adapters	UD / UW P1
Mutation through adapters (incl. post-verification action drift)	UD / UW P1
Adversarial malformed input	UD / UW P1
ProofRecord tamper / forgery	UD
Forged, stale, reused, or revoked evidence/authority	UD / UW P1
False-ALLOW hunting / containment-resistance testing	UD / UW P1
Independent reproduction claims	UW P4
External conformance claims	UW P4

15. Prior-work disclosure (summary — full text in `PRIOR_WORK_DISCLOSURE.md`)

RF has prior, RF-run, implementation-specific work on the EP boundary, including the ARK corpus (ARK-493–498, 10.5281/zenodo.21462612; P01 series ARK-458–482), the intent-binding series IB-001–003 (10.5281/zenodo.22002829; IB-001/002 FAIL preserved, IB-003 PASS), authority partitioning AUTH-001/002 (10.5281/zenodo.22002833), and `PRODUCTION-STATUS.md` (executionproof v1.0.0-public). EP-AGENT-001 is intentionally separated from all of it: it imports no adversarial case tables, reuses no ARK/EP-SEC cases, and answers a different question (integration portability).

16. Preregistration freeze checklist

All 14 items must be checked, with evidence, before the SHA-256 lock is computed and before any execution authorization can be requested.

#	Item	Status
1	Primary RQ, CQ separation, and hypotheses H1–H4 reviewed and frozen	✔ PASS — consistency audit v0.3: RQ/CQ/H1–H4 unchanged
2	Primary endpoints P1–P5 formulas and thresholds frozen	✔ PASS — P1–P5 formulas/thresholds identical across PREREGISTRATION, ANALYSIS_PLAN, PASS_FAIL_RULES
3	Fixtures B1–B5 frozen; SHA-256 of each fixture recorded	✔ PASS — B1–B5 hashes in MANIFEST (unchanged since v0.2)
4	`EXPECTED_RESULTS.json` frozen; SHA-256 recorded	✔ PASS — sha256 af7ea17a…72d9; verdicts match fixtures
5	Request Contract schema (`REQUEST_CONTRACT.md`) frozen; normalization rules per framework frozen	✔ PASS — 20/20 normalized contracts schema-valid
6		

#	Item	Status
	Framework versions pinned in `requirements.lock`; lockfile SHA-256 recorded	✔ PASS — requirements.lock + locks/.lock verified in clean venvs
7	ExecutionProof version/commit pinned; fixture policy configuration hash recorded	✔ PASS — v0.3.3: `v0.1.2-reference` @ `b03365d5…`; policy `06b5a7ab…77649d4` (unchanged); evidence registry `b4a84f29…490747d0` (Derek Hone option a); load-only `/health` check only
8	Adapter code implemented (stubs replaced), reviewed for scope (no delegation, no bypass/replay/mutation paths), commit pinned	✔ PASS — real adapters + bindings; single-agent/single-tool asserted in smoke
9	Dry-run on a non-study smoke fixture completed (harness only; not B1–B5) and logged as a dry-run, not a result	✔ PASS — SMOKE-000 engineering smoke 4/4 (ENGINEERING_SMOKE_LOG.md)
10	Analysis scripts, LOC counting rule, latency protocol, and bootstrap seed frozen	✔ PASS — analyze.py + synthetic self-test 14/14; seed 20261015
11	PASS / PARTIAL / FAIL / INCONCLUSIVE rules and invalidation conditions frozen	✔ PASS — PASS_FAIL_RULES v0.3 precedence/anti-rescue
12	10-question university non-interference gate re-run and recorded (design-scope verdict GO or GO WITH BOUNDARY **and** Q8, Q9 PASSED)	☐ OPEN — Q9 PASSED (v0.3.1); Q8 technical readiness PASS (v0.3.3); Q8 formal not PASSED (awaiting Derek approval + filing)
13	`UNIVERSITY_EXPOSURE_LOG.md` reconciled (historical UD entries resolved; UW assignments checked)	✔ PASS — v0.3.1: Gmail record search (355 msgs, 46 attachments) + Derek Hone written confirmation (§5b)
14	`MANIFEST.sha256` regenerated and verified (`sha256sum -c`); git commit hash and UTC timestamp recorded; Derek Hone's written approval of the packet recorded; preregistration filing record created (no digital signature is part of the lock)	☐ OPEN

17. Authorization

Execution requires, in order: (1) completion of the freeze checklist, (2) a SHA-locked preregistration, (3) a passing new-research gate (Q8 and Q9 PASSED), and (4) Derek Hone's explicit written execution authorization (template in `EP-AGENT-001_DESIGN_SUMMARY.md`, Section Z). None of these has occurred.

18. PENDING AMENDMENT — DRAFT, NOT FILED

This amendment is a draft. It has not been approved by Derek Hone and has not been filed. Until it is approved and filed, the sections above remain the operative draft text. (The header status lines were updated in v0.3.3 and now record item 7 PASS and Q8 technical freeze readiness PASS; superseded v0.3.2 wording: "the header status lines, which still record item 7 as blocked".)

ExecutionProof under test (verbatim):

> "ExecutionProof under test is a controlled reference implementation built by Remnant Fieldworks before preregistration filing and before experimental execution. The implementation was frozen by repository commit and configuration hashes before the EP-AGENT-001 experimental matrix was run. The study therefore evaluates portability across selected agent frameworks against this reference implementation and does not establish performance of a separately deployed production ExecutionProof system."

Candidate pin: ExecutionProof Controlled Reference Verifier, tag `v0.1.2-reference`, commit `b03365d57880e896c35e72898f8335a1826afe74`, loaded with study policy `fixtures/policy/fixture_policy_config.json` (SHA-256 `06b5a7ab68696d4d246f7b07f698437ec5713fa1da1113f553a02e04777649d4`) via `EP_POLICY_CONFIG` and study evidence registry `fixtures/evidence/study_evidence_registry.json` (SHA-256 `b4a84f293182cd299b9168`

f3d38f3fc8f00e56c6bc89bcfa74a6385a490747d0) via EP_EVIDENCE_REGISTRY (details and hashes in EXECUTIONPROOF_PIN.md). Evidence-reference status mechanism: Derek Hone option (a), separate hashed fail-closed registry. Checklist item 7: PASS (v0.3.3). Superseded v0.3.2 text: candidate v0.1.1-reference @ 279bcb2669e4602ae25e0d745d827a1c8514265e, item 7 OPEN.

Narrowed claim ceiling (applies if this amendment is adopted): the §13 maximum claim refers only to the /verify decision path of the pinned ExecutionProof Controlled Reference Verifier (tag v0.1.2-reference, commit b03365d57880e896c35e72898f8335a1826afe74, with the study policy and study evidence registry hashes above) and establishes portability against that pinned reference implementation only; it may not be broadened to any other ExecutionProof build, deployment, or version.

SHA-256 of this document: final value recorded at filing. Current value in MANIFEST.sha256 = PRE-FILING CANDIDATE LOCK.